

Unit 4. Partitioned Matrices and the Schur Complement

1 Block factors

1.1 Block-triangular forms

Definition 4.1 (Block diagonal and block triangular). Let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \in \mathbb{R}^{(m+n) \times (m+n)}, \quad \mathbf{A} \in \mathbb{R}^{m \times m}, \quad \mathbf{D} \in \mathbb{R}^{n \times n}.$$

It is **block diagonal**, **upper block triangular**, or **lower block triangular** when it has, respectively, one of the forms

$$\begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}, \quad \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}, \quad \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}.$$

Origin: H §2.1; A&M Ex. 5.5

Theorem 4.2 (Unit block factors). For $\mathbf{X}, \mathbf{Z} \in \mathbb{R}^{m \times n}$ and $\mathbf{Y}, \mathbf{W} \in \mathbb{R}^{n \times m}$, define

$$\mathbf{U}(\mathbf{X}) = \begin{pmatrix} \mathbf{I}_m & \mathbf{X} \\ \mathbf{0} & \mathbf{I}_n \end{pmatrix}, \quad \mathbf{L}(\mathbf{Y}) = \begin{pmatrix} \mathbf{I}_m & \mathbf{0} \\ \mathbf{Y} & \mathbf{I}_n \end{pmatrix}.$$

Then

$$\mathbf{U}(\mathbf{X})^{-1} = \mathbf{U}(-\mathbf{X}), \quad \mathbf{L}(\mathbf{Y})^{-1} = \mathbf{L}(-\mathbf{Y}), \quad \det \mathbf{U}(\mathbf{X}) = \det \mathbf{L}(\mathbf{Y}) = 1,$$

and

$$\mathbf{U}(\mathbf{X})\mathbf{U}(\mathbf{Z}) = \mathbf{U}(\mathbf{X} + \mathbf{Z}), \quad \mathbf{L}(\mathbf{Y})\mathbf{L}(\mathbf{W}) = \mathbf{L}(\mathbf{Y} + \mathbf{W}).$$

Origin: H Lemma 8.5.2, Eqs. 8.5.6a–b; A&M Ex. 5.9 and 5.13(a)

Proof. Block multiplication gives

$$\mathbf{U}(\mathbf{X})\mathbf{U}(-\mathbf{X}) = \begin{pmatrix} \mathbf{I}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_n \end{pmatrix}, \quad \mathbf{U}(\mathbf{X})\mathbf{U}(\mathbf{Z}) = \begin{pmatrix} \mathbf{I}_m & \mathbf{X} + \mathbf{Z} \\ \mathbf{0} & \mathbf{I}_n \end{pmatrix}.$$

The lower-block identities follow with $\mathbf{Y}, \mathbf{W}$ in place of $\mathbf{X}, \mathbf{Z}$. Both matrices are triangular with every diagonal entry equal to 1; hence both determinants equal 1. $\square$

1.2 Block-triangular algebra

Theorem 4.3 (Block-triangular determinant and inverse). Let $\mathbf{A} \in \mathbb{R}^{m \times m}$, $\mathbf{D} \in \mathbb{R}^{n \times n}$, $\mathbf{B} \in \mathbb{R}^{m \times n}$, and $\mathbf{C} \in \mathbb{R}^{n \times m}$. Then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix} = \det \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A}) \det(\mathbf{D}).$$

Either block-triangular matrix is nonsingular if and only if both $\mathbf{A}$ and $\mathbf{D}$ are nonsingular. In that case,

$$\begin{aligned} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}^{-1} &= \begin{pmatrix} \mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{D}^{-1} \end{pmatrix}, \\ \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^{-1} &= \begin{pmatrix} \mathbf{A}^{-1} & \mathbf{O} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{D}^{-1} \end{pmatrix}. \end{aligned}$$

Moreover,

$$\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{D} \end{pmatrix} = \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{D}).$$

Origin: H Lemma 8.5.4, Thm. 13.3.1, and Lemma 4.5.11; A&M Ex. 5.13, 5.25, and 5.41(a)

Proof. Set $N = m + n$ and $\mathbf{T} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}$. In the permutation formula for $\det(\mathbf{T})$,

$$\prod_{i=1}^N t_{i,\sigma(i)} \neq 0 \implies \{\sigma(m+1), \dots, \sigma(N)\} = \{m+1, \dots, N\}.$$

Thus there are unique $\sigma_A \in S_m$ and $\sigma_D \in S_n$ such that

$$\sigma(i) = \sigma_A(i), \quad \sigma(m+j) = m + \sigma_D(j).$$

There are no cross-block inversions, so $\text{sgn}(\sigma) = \text{sgn}(\sigma_A) \text{sgn}(\sigma_D)$. Hence

$$\det(\mathbf{T}) = \sum_{\sigma_A \in S_m} \text{sgn}(\sigma_A) \prod_{i=1}^m a_{i, \sigma_A(i)} \sum_{\sigma_D \in S_n} \text{sgn}(\sigma_D) \prod_{j=1}^n d_{j, \sigma_D(j)} = \det(\mathbf{A}) \det(\mathbf{D}).$$

Transpose invariance gives the lower-block formula. The determinant criterion gives

$$\mathbf{T} \text{ nonsingular} \iff \det(\mathbf{A}) \det(\mathbf{D}) \neq 0 \iff \mathbf{A}, \mathbf{D} \text{ nonsingular.}$$

Direct multiplication of each matrix by its displayed candidate inverse gives $\mathbf{I}_{m+n}$.

For $\mathbf{K} = \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{D} \end{pmatrix},$

$$\text{col}(\mathbf{K}) = \left\{ \begin{pmatrix} \mathbf{u} \\ \mathbf{0} \end{pmatrix} : \mathbf{u} \in \text{col}(\mathbf{A}) \right\} \oplus \left\{ \begin{pmatrix} \mathbf{0} \\ \mathbf{v} \end{pmatrix} : \mathbf{v} \in \text{col}(\mathbf{D}) \right\}.$$

Taking dimensions proves the rank identity. □

Example 4.1 (An exact block-triangular inverse). With a $2 + 1$ partition,

$$\mathbf{T} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Since

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad -\mathbf{A}^{-1}\mathbf{B}\mathbf{D}^{-1} = \begin{pmatrix} -1 \\ 1/2 \end{pmatrix},$$

$$\mathbf{T}^{-1} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1/2 \end{pmatrix}, \quad \det(\mathbf{T}) = 2.$$

Origin: Course formulation from H Lemma 8.5.4 and Thm. 13.3.1

1.3 Schur complements and block factorisation

Definition 4.4 (Schur complements). For

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}, \quad \mathbf{A} \in \mathbb{R}^{m \times m}, \quad \mathbf{D} \in \mathbb{R}^{n \times n},$$

the Schur complement of $\mathbf{A}$ in $\mathbf{M}$, when $\mathbf{A}$ is nonsingular, is

$$\mathbf{S}_A = \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}.$$

When $\mathbf{D}$ is nonsingular, the Schur complement of $\mathbf{D}$ in $\mathbf{M}$ is

$$\mathbf{S}_D = \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}.$$

Origin: H §8.5e after Eq. 8.5.18; A&M Ex. 5.10–5.11; S Def. 3.3

Theorem 4.5 (Schur factorisations). If $\mathbf{A}$ is nonsingular, then

$$\begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ -\mathbf{CA}^{-1} & \mathbf{I}_n \end{pmatrix} \mathbf{M} \begin{pmatrix} \mathbf{I}_m & -\mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_A \end{pmatrix}, \quad (1)$$

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{CA}^{-1} & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_A \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix}. \quad (2)$$

If $\mathbf{D}$ is nonsingular, then

$$\begin{pmatrix} \mathbf{I}_m & -\mathbf{BD}^{-1} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix} \mathbf{M} \begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ -\mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_n \end{pmatrix} = \begin{pmatrix} \mathbf{S}_D & \mathbf{O} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}, \quad (3)$$

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_m & \mathbf{BD}^{-1} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{S}_D & \mathbf{O} \\ \mathbf{O} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_n \end{pmatrix}. \quad (4)$$

Origin: H Eq. 8.5.18 and Thm. 13.3.8; A&M Ex. 5.10–5.11; S Prop. 3.9

Proof. Using the nonsingular block $\mathbf{A}$,

$$\begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ -\mathbf{CA}^{-1} & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{S}_A \end{pmatrix},$$

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{S}_A \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & -\mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_A \end{pmatrix}.$$

Multiplication by the inverses from Theorem 4.2 gives (2). Using the nonsingular block $\mathbf{D}$,

$$\begin{pmatrix} \mathbf{I}_m & -\mathbf{BD}^{-1} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{S}_D & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{pmatrix},$$

$$\begin{pmatrix} \mathbf{S}_D & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ -\mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_n \end{pmatrix} = \begin{pmatrix} \mathbf{S}_D & \mathbf{O} \\ \mathbf{O} & \mathbf{D} \end{pmatrix}.$$

Inverting the two unit factors gives (4). □

Example 4.2 (Exact Schur factors). Let

$$\mathbf{M} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 3 & 1 & 4 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{C} = (3 \ 1), \quad \mathbf{D} = (4).$$

Then

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{A}^{-1}\mathbf{B} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}, \quad \mathbf{C}\mathbf{A}^{-1} = (2 \ -1), \quad \mathbf{S}_A = (4).$$

Thus

$$\mathbf{M} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Origin: Course formulation from A&M Ex. 5.10; H Eq. 8.5.18

2 Schur consequences

2.1 Rank and nonsingularity

Theorem 4.6 (Schur rank and nonsingularity). Let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}, \quad \mathbf{A} \in \mathbb{R}^{m \times m}, \quad \mathbf{D} \in \mathbb{R}^{n \times n}.$$

If $\mathbf{A}$ is nonsingular, then

$$\text{rank}(\mathbf{M}) = m + \text{rank}(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}).$$

Moreover,

$$\mathbf{M} \text{ is nonsingular} \iff \mathbf{S}_A \text{ is nonsingular.}$$

If $\mathbf{D}$ is nonsingular,

$$\text{rank}(\mathbf{M}) = n + \text{rank}(\mathbf{S}_D), \quad \mathbf{M} \text{ is nonsingular} \iff \mathbf{S}_D \text{ is nonsingular.}$$

Origin: H Thm. 8.5.10–8.5.11, Eq. 8.5.15; A&M Ex. 5.48

Proof. The two unit factors in (1) are nonsingular. Rank is invariant under multiplication by nonsingular matrices, so

$$\text{rank}(\mathbf{M}) = \text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_A \end{pmatrix} = \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{S}_A) = m + \text{rank}(\mathbf{S}_A).$$

$$\text{rank}(\mathbf{M}) = m + n \iff \text{rank}(\mathbf{S}_A) = n \iff \mathbf{S}_A \text{ is nonsingular.}$$

The **D** formulas follow from (3). □

Example 4.3 (A null direction exposed by the complement). For

$$\mathbf{M} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 3 & 4 & 11 \end{pmatrix} = \begin{pmatrix} \mathbf{I}_2 & \begin{matrix} 1 \\ 2 \end{matrix} \\ 3 & 4 & 11 \end{pmatrix},$$

$$\mathbf{S}_A = 11 - (3 \ 4) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 0, \quad \text{rank}(\mathbf{M}) = 2, \quad \mathbf{M} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} = \mathbf{0}.$$

Origin: Course formulation from H Thm. 8.5.10

2.2 The block inverse

Theorem 4.7 (Block inverse). If $\mathbf{A}$ and $\mathbf{S}_A$ are nonsingular, then

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{S}_A^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}_A^{-1} \\ -\mathbf{S}_A^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{S}_A^{-1} \end{pmatrix}. \quad (5)$$

If $\mathbf{D}$ and $\mathbf{S}_D$ are nonsingular, then

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{S}_D^{-1} & -\mathbf{S}_D^{-1}\mathbf{B}\mathbf{D}^{-1} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{S}_D^{-1} & \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{S}_D^{-1}\mathbf{B}\mathbf{D}^{-1} \end{pmatrix}. \quad (6)$$

Origin: H Thm. 8.5.11, Eqs. 8.5.16–8.5.18; A&M Ex. 5.16

Proof. Invert the three factors in (2):

$$\begin{aligned}\mathbf{M}^{-1} &= \begin{pmatrix} \mathbf{I}_m & -\mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{A}^{-1} & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_A^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ -\mathbf{C}\mathbf{A}^{-1} & \mathbf{I}_n \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{S}_A^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}_A^{-1} \\ -\mathbf{S}_A^{-1}\mathbf{C}\mathbf{A}^{-1} & \mathbf{S}_A^{-1} \end{pmatrix}.\end{aligned}$$

Likewise, (4) gives

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ -\mathbf{D}^{-1}\mathbf{C} & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{S}_D^{-1} & \mathbf{O} \\ \mathbf{O} & \mathbf{D}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & -\mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix},$$

whose product is (6). □

Corollary 4.8 (A diagonal inverse block determines the complement). Suppose $\mathbf{M}$ and $\mathbf{A}$ are nonsingular and write

$$\mathbf{M}^{-1} = \begin{pmatrix} \mathbf{P} & \mathbf{Q} \\ \mathbf{R} & \mathbf{T} \end{pmatrix}.$$

Then $\mathbf{T}$ is nonsingular and

$$\begin{aligned} \mathbf{S}_A = \mathbf{T}^{-1}, \quad \mathbf{A}^{-1} &= \mathbf{P} - \mathbf{Q}\mathbf{T}^{-1}\mathbf{R}, \\ \mathbf{A}^{-1}\mathbf{B} &= -\mathbf{Q}\mathbf{T}^{-1}, \quad \mathbf{C}\mathbf{A}^{-1} = -\mathbf{T}^{-1}\mathbf{R}. \end{aligned}$$

Origin: H Cor. 8.5.12, Eqs. 8.5.19–8.5.24; A&M Ex. 5.17

Proof. Theorem 4.6 makes $\mathbf{S}_A$ nonsingular. Comparing blocks in (5) gives

$$\mathbf{T} = \mathbf{S}_A^{-1}, \quad \mathbf{Q} = -\mathbf{A}^{-1}\mathbf{B}\mathbf{T}, \quad \mathbf{R} = -\mathbf{T}\mathbf{C}\mathbf{A}^{-1},$$

and

$$\mathbf{P} - \mathbf{Q}\mathbf{T}^{-1}\mathbf{R} = \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\mathbf{T}\mathbf{C}\mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}\mathbf{T}\mathbf{C}\mathbf{A}^{-1} = \mathbf{A}^{-1}.$$

□

Example 4.4 (The inverse from one scalar complement). For the matrix in Example 4.2, $\mathbf{S}_A = (4)$, and

$$\mathbf{M}^{-1} = \begin{pmatrix} 1/2 & -3/4 & 1/4 \\ 1/2 & 5/4 & -3/4 \\ -1/2 & 1/4 & 1/4 \end{pmatrix}.$$

Its lower-right entry is $\mathbf{S}_A^{-1} = 1/4$, and

$$\mathbf{M}\mathbf{M}^{-1} = \mathbf{I}_3.$$

Origin: Course formulation from H Thm. 8.5.11; A&M Ex. 5.16

2.3 Determinants on the smaller block

Theorem 4.9 (Schur determinant identities). For square $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$,

$$\det(\mathbf{M}) = \det(\mathbf{A}) \det(\mathbf{S}_A) \quad \text{if } \mathbf{A} \text{ is nonsingular,}$$

and

$$\det(\mathbf{M}) = \det(\mathbf{D}) \det(\mathbf{S}_D) \quad \text{if } \mathbf{D} \text{ is nonsingular.}$$

Origin: H Thm. 13.3.8, Eq. 13.3.13; A&M Ex. 5.30; S Prop. 3.9

Proof. Determinant multiplicativity, Theorem 4.2, and (2) give

$$\begin{aligned} \det(\mathbf{M}) &= \det \begin{pmatrix} \mathbf{I}_m & \mathbf{O} \\ \mathbf{C}\mathbf{A}^{-1} & \mathbf{I}_n \end{pmatrix} \det \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_A \end{pmatrix} \det \begin{pmatrix} \mathbf{I}_m & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O} & \mathbf{I}_n \end{pmatrix} \\ &= 1 \cdot \det(\mathbf{A}) \det(\mathbf{S}_A) \cdot 1. \end{aligned}$$

The same calculation applied to (4) gives the second identity. □

Corollary 4.10 (Sylvester's determinant identity). For $\mathbf{U} \in \mathbb{R}^{m \times k}$ and $\mathbf{V} \in \mathbb{R}^{k \times m}$,

$$\det(\mathbf{I}_m + \mathbf{U}\mathbf{V}) = \det(\mathbf{I}_k + \mathbf{V}\mathbf{U}).$$

Consequently,

$$\mathbf{I}_m + \mathbf{UV} \text{ is nonsingular} \iff \mathbf{I}_k + \mathbf{VU} \text{ is nonsingular.}$$

Origin: H Cor. 18.1.2, Eq. 18.1.2; A&M Ex. 5.37(b)

Proof. Apply the two identities of Theorem 4.9 to

$$\mathbf{K} = \begin{pmatrix} \mathbf{I}_m & \mathbf{U} \\ -\mathbf{V} & \mathbf{I}_k \end{pmatrix}.$$

The two complements are $\mathbf{I}_k + \mathbf{VU}$ and $\mathbf{I}_m + \mathbf{UV}$; hence

$$\det(\mathbf{K}) = \det(\mathbf{I}_k + \mathbf{VU}) = \det(\mathbf{I}_m + \mathbf{UV}).$$

The determinant criterion proves the equivalence. □

Example 4.5 (Two determinant orders). Let

$$\mathbf{U} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Then

$$\mathbf{I}_3 + \mathbf{UV} = \begin{pmatrix} 2 & 2 & 0 \\ 0 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix}, \quad \mathbf{I}_2 + \mathbf{VU} = \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix},$$

and both determinants equal 4.

Origin: Course formulation from H Cor. 18.1.2; A&M Ex. 5.37(b)

3 Determinant and inverse updates

3.1 A small determinant update

Theorem 4.11 (General determinant update). Let

$$\mathbf{A} \in \mathbb{R}^{n \times n}, \quad \mathbf{U} \in \mathbb{R}^{n \times k}, \quad \mathbf{C} \in \mathbb{R}^{k \times k}, \quad \mathbf{V} \in \mathbb{R}^{k \times n},$$

where $\mathbf{A}$ and $\mathbf{C}$ are nonsingular. Then

$$\det(\mathbf{A} + \mathbf{UCV}) = \det(\mathbf{A}) \det(\mathbf{C}) \det(\mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U}).$$

Origin: H Thm. 18.1.1, Eq. 18.1.1; A&M Ex. 5.37(c)

Proof. By multiplicativity and Corollary 4.10,

$$\begin{aligned}\det(\mathbf{A} + \mathbf{UCV}) &= \det(\mathbf{A}) \det(\mathbf{I}_n + \mathbf{A}^{-1}\mathbf{UCV}) \\ &= \det(\mathbf{A}) \det(\mathbf{I}_k + \mathbf{VA}^{-1}\mathbf{UC}) \\ &= \det(\mathbf{A}) \det((\mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U})\mathbf{C}) \\ &= \det(\mathbf{A}) \det(\mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U}) \det(\mathbf{C}).\end{aligned}$$

□

3.2 Woodbury's identity

Theorem 4.12 (Woodbury identity). Under the hypotheses of Theorem 4.11,

$$\mathbf{A} + \mathbf{UCV} \text{ is nonsingular} \iff \mathbf{E} := \mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U} \text{ is nonsingular.}$$

When these conditions hold,

$$(\mathbf{A} + \mathbf{UCV})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{UE}^{-1}\mathbf{VA}^{-1}.$$

Origin: H Thm. 18.2.8, Eq. 18.2.22; A&M Ex. 5.17

Proof. The equivalence follows from Theorem 4.11 and $\det(\mathbf{A})\det(\mathbf{C}) \neq 0$. For the inverse, set

$$\mathbf{K} = \begin{pmatrix} \mathbf{A} & \mathbf{U} \\ -\mathbf{V} & \mathbf{C}^{-1} \end{pmatrix}.$$

Its complements formed from $\mathbf{A}$ and $\mathbf{C}^{-1}$ are

$$\mathbf{C}^{-1} - (-\mathbf{V})\mathbf{A}^{-1}\mathbf{U} = \mathbf{E}, \quad \mathbf{A} - \mathbf{UC}(-\mathbf{V}) = \mathbf{A} + \mathbf{UCV}.$$

The upper-left block of (5) is

$$\mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\mathbf{E}^{-1}\mathbf{V}\mathbf{A}^{-1},$$

whereas the upper-left block of (6) is $(\mathbf{A} + \mathbf{UCV})^{-1}$. □

Corollary 4.13 (Push-through identities). For $\mathbf{U} \in \mathbb{R}^{m \times k}$ and $\mathbf{V} \in \mathbb{R}^{k \times m}$, if $\mathbf{I}_k + \mathbf{VU}$ is nonsingular, then

$$(\mathbf{I}_m + \mathbf{UV})^{-1} = \mathbf{I}_m - \mathbf{U}(\mathbf{I}_k + \mathbf{VU})^{-1}\mathbf{V},$$

and

$$(\mathbf{I}_m + \mathbf{UV})^{-1}\mathbf{U} = \mathbf{U}(\mathbf{I}_k + \mathbf{VU})^{-1}.$$

Origin: H Cor. 18.2.9, Eq. 18.2.24

Proof. The first identity is Theorem 4.12 with both diagonal matrices equal to identities. Put $\mathbf{K} = \mathbf{I}_k + \mathbf{VU}$. Then

$$\begin{aligned} [\mathbf{I}_m - \mathbf{UK}^{-1}\mathbf{V}]\mathbf{U} &= \mathbf{U}[\mathbf{I}_k - \mathbf{K}^{-1}\mathbf{VU}] \\ &= \mathbf{UK}^{-1}[\mathbf{K} - \mathbf{VU}] = \mathbf{UK}^{-1}. \end{aligned} \quad \square$$

3.3 Rank-one updates and borders

Corollary 4.14 (Sherman–Morrison identity). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be nonsingular and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Then

$$\mathbf{A} + \mathbf{u}\mathbf{v}' \text{ is nonsingular} \iff 1 + \mathbf{v}'\mathbf{A}^{-1}\mathbf{u} \neq 0.$$

Under this condition,

$$(\mathbf{A} + \mathbf{u}\mathbf{v}')^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}'\mathbf{A}^{-1}}{1 + \mathbf{v}'\mathbf{A}^{-1}\mathbf{u}}.$$

Origin: H Cor. 18.2.10, Eq. 18.2.25; S Prop. 3.21; A&M Ex. 4.28

Proof. Take $k = 1$, $\mathbf{U} = \mathbf{u}$, $\mathbf{C} = (1)$, and $\mathbf{V} = \mathbf{v}'$ in Theorem 4.12. The matrix $\mathbf{E}$ is the scalar $1 + \mathbf{v}'\mathbf{A}^{-1}\mathbf{u}$. □

Corollary 4.15 (A scalar border). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be nonsingular, $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and $\delta \in \mathbb{R}$. Put

$$\varepsilon = \delta - \mathbf{x}'\mathbf{A}^{-1}\mathbf{y}, \quad \mathbf{K} = \begin{pmatrix} \mathbf{A} & \mathbf{y} \\ \mathbf{x}' & \delta \end{pmatrix}.$$

Then

$$\det(\mathbf{K}) = \det(\mathbf{A})\varepsilon, \quad \mathbf{K} \text{ is nonsingular} \iff \varepsilon \neq 0.$$

If $\varepsilon \neq 0$, then

$$\mathbf{K}^{-1} = \begin{pmatrix} \mathbf{A}^{-1} + \varepsilon^{-1}\mathbf{A}^{-1}\mathbf{y}\mathbf{x}'\mathbf{A}^{-1} & -\varepsilon^{-1}\mathbf{A}^{-1}\mathbf{y} \\ -\varepsilon^{-1}\mathbf{x}'\mathbf{A}^{-1} & \varepsilon^{-1} \end{pmatrix}.$$

Origin: H Thm. 8.5.11 and Ex. 13.12; A&M Ex. 5.15 and 5.29

Proof. Here $\mathbf{S}_A = (\varepsilon)$. Apply Theorems 4.6, 4.7, and 4.9. □

Example 4.6 (An exact rank-one update). Let

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Then

$$1 + \mathbf{v}'\mathbf{A}^{-1}\mathbf{u} = \frac{1}{2}, \quad \mathbf{A} + \mathbf{u}\mathbf{v}' = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix},$$

and

$$(\mathbf{A} + \mathbf{u}\mathbf{v}')^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1 \end{pmatrix} - 2 \begin{pmatrix} 1/2 \\ 1 \end{pmatrix} (1/2 \quad -1) = \begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}.$$

Origin: Course formulation from H Cor. 18.2.10; S Prop. 3.21

4 Computation: Numerical Checks in R

4.1 Block factorisation (§1.3)

```
1 A <- matrix(c(2, 1,  
2           1, 1), 2, 2, byrow = TRUE)  
3 B <- matrix(c(1, 2), 2, 1)  
4 C <- matrix(c(3, 1), 1, 2)  
5 D <- matrix(4, 1, 1)  
6 Ainv <- solve(A)  
7 S <- D - C %*% Ainv %*% B  
8 M <- rbind(cbind(A, B), cbind(C, D))
```

```

9 L <- rbind(cbind(diag(2), matrix(0, 2, 1)),
10           cbind(C %*% Ainv, diag(1)))
11 Delta <- rbind(cbind(A, matrix(0, 2, 1)),
12              cbind(matrix(0, 1, 2), S))
13 Ufac <- rbind(cbind(diag(2), Ainv %*% B),
14              cbind(matrix(0, 1, 2), diag(1)))
15 c(schur_complement = S[1, 1],
16   reconstruction_residual = max(abs(M - L %*% Delta %*% Ufac)))

```

```

##           schur_complement reconstruction_residual
##                4                        0

```

4.2 Block inverse and determinant (§2.2, §2.3)

```

1 Sinv <- solve(S)
2 manual_inverse <- rbind(
3   cbind(Ainv + Ainv %*% B %*% Sinv %*% C %*% Ainv,
4         -Ainv %*% B %*% Sinv),
5   cbind(-Sinv %*% C %*% Ainv, Sinv)
6 )
7 package_inverse <- solve(M)
8 c(inverse_difference = max(abs(manual_inverse - package_inverse)),

```

```
9   inverse_residual = max(abs(M %*% manual_inverse - diag(3))),  
10  determinant_difference = det(M) - det(A) * det(S))
```

```
##      inverse_difference      inverse_residual  
##      2.220e-16              0.000e+00  
## determinant_difference  
##      8.882e-16
```

4.3 The smaller determinant (§2.3)

```
1  set.seed(4)  
2  U <- matrix(rnorm(160), 80, 2)  
3  V <- matrix(rnorm(160), 2, 80) / 80  
4  large_det <- det(diag(80) + U %*% V)  
5  small_det <- det(diag(2) + V %*% U)  
6  c(order_80 = large_det,  
7    order_2 = small_det,  
8    difference = large_det - small_det)
```

```
##      order_80      order_2 difference  
## 9.627e-01 9.627e-01 7.772e-16
```

4.4 Woodbury against solve (§3.2)

```
1  set.seed(44)
2  n <- 12
3  k <- 2
4  A <- diag(seq(2, 3, length.out = n))
5  C <- matrix(c(2, 0.25,
6               0.25, 1.5), 2, 2, byrow = TRUE)
7  U <- matrix(rnorm(n * k), n, k)
8  V <- matrix(rnorm(k * n), k, n) / n
9  Ainv <- solve(A)
10 E <- solve(C) + V %*% Ainv %*% U
11 manual_inverse <- Ainv - Ainv %*% U %*% solve(E) %*% V %*% Ainv
12 updated <- A + U %*% C %*% V
13 package_inverse <- solve(updated)
14 direct_det <- det(updated)
15 update_det <- det(A) * det(C) * det(E)
16 c(inverse_difference = max(abs(manual_inverse - package_inverse)),
17   inverse_residual = max(abs(updated %*% manual_inverse - diag(n))),
18   determinant_relative_difference =
```

```
19 (direct_det - update_det) / max(1, abs(direct_det))
```

```
##          inverse_difference
##          1.110e-16
##          inverse_residual
##          2.220e-16
## determinant_relative_difference
##          -1.169e-15
```

5 Pitfalls

5.1 Using the scalar 2 by 2 determinant rule

For

$$\mathbf{A} = \mathbf{D} = \mathbf{I}_2, \quad \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{I}_2 - \mathbf{CB}) = -1, \quad \det(\mathbf{A}) \det(\mathbf{D}) - \det(\mathbf{B}) \det(\mathbf{C}) = 0.$$

5.2 Requiring a nonsingular diagonal block

Both diagonal blocks of

$$\begin{pmatrix} \mathbf{O} & \mathbf{I}_n \\ \mathbf{I}_n & \mathbf{O} \end{pmatrix}$$

are singular, but the matrix is its own inverse. A Schur formula based on one diagonal block does not characterize all nonsingular block matrices.

5.3 Inverting each displayed block

For nonsingular $\mathbf{B}$,

$$\begin{pmatrix} \mathbf{I} & \mathbf{B} \\ \mathbf{O} & \mathbf{I} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{B} \\ \mathbf{O} & \mathbf{I} \end{pmatrix} \neq \begin{pmatrix} \mathbf{I} & \mathbf{B}^{-1} \\ \mathbf{O} & \mathbf{I} \end{pmatrix}.$$

5.4 Ignoring the update denominator

For $\mathbf{A} = \mathbf{I}_n$, $\mathbf{u} = \mathbf{e}_1$, and $\mathbf{v} = -\mathbf{e}_1$,

$$1 + \mathbf{v}'\mathbf{A}^{-1}\mathbf{u} = 0, \quad \mathbf{A} + \mathbf{u}\mathbf{v}' = \text{diag}(0, 1, \dots, 1).$$

The zero denominator in Corollary 4.14 is exactly the singular case.

6 Exercises

Exercise 4.1 (Unit block factors and positive powers). For $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{m \times n}$ and $r \in \{1, 2, \dots\}$, prove

$$\mathbf{U}(\mathbf{X})\mathbf{U}(\mathbf{Y}) = \mathbf{U}(\mathbf{Y})\mathbf{U}(\mathbf{X}) = \mathbf{U}(\mathbf{X} + \mathbf{Y}), \quad \mathbf{U}(\mathbf{X})^r = \mathbf{U}(r\mathbf{X}).$$

Origin: H Lemma 8.5.2; A&M Ex. 5.9 and 5.21

Solution. Theorem 4.2 gives the product identity. The power formula holds for $r = 1$. If it holds for $r \geq 1$, then

$$\mathbf{U}(\mathbf{X})^{r+1} = \mathbf{U}(r\mathbf{X})\mathbf{U}(\mathbf{X}) = \mathbf{U}((r+1)\mathbf{X}).$$



Exercise 4.2 (A partitioned linear system). Let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}, \quad \mathbf{S}_A = \mathbf{D} - \mathbf{CA}^{-1}\mathbf{B},$$

where $\mathbf{A}$ and $\mathbf{S}_A$ are nonsingular. For conformable $\mathbf{f}, \mathbf{g}$, prove that the unique solution of

$$\mathbf{M} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{f} \\ \mathbf{g} \end{pmatrix}$$

is

$$\mathbf{y} = \mathbf{S}_A^{-1}(\mathbf{g} - \mathbf{CA}^{-1}\mathbf{f}), \quad \mathbf{x} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{By}).$$

Origin: Course formulation from H Thm. 8.5.11; A&M Ex. 5.16

Solution. The first block equation gives

$$\mathbf{x} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{By}).$$

Substitution in the second gives

$$\mathbf{CA}^{-1}(\mathbf{f} - \mathbf{By}) + \mathbf{Dy} = \mathbf{g} \iff \mathbf{S}_A\mathbf{y} = \mathbf{g} - \mathbf{CA}^{-1}\mathbf{f}.$$

Nonsingularity of $\mathbf{S}_A$ gives the displayed $\mathbf{y}$, then the displayed $\mathbf{x}$. Theorem 4.6 gives nonsingularity of the coefficient matrix, so the solution is unique. 

Exercise 4.3 (Off-diagonal coupling). Let $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{n \times n}$ be nonsingular and

$$\mathbf{K} = \begin{pmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{C} & \mathbf{0} \end{pmatrix}.$$

Find $\mathbf{K}^{-1}$, $\det(\mathbf{K})$, and $\text{rank}(\mathbf{K})$.

Origin: H Cor. 13.3.2; A&M Ex. 5.12(b), 5.23, and 5.41(b)

Solution.

$$\mathbf{K} \begin{pmatrix} \mathbf{0} & \mathbf{C}^{-1} \\ \mathbf{B}^{-1} & \mathbf{0} \end{pmatrix} = \mathbf{I}_{2n}, \quad \text{rank}(\mathbf{K}) = 2n.$$

Let $\mathbf{P} = \begin{pmatrix} \mathbf{0} & \mathbf{I}_n \\ \mathbf{I}_n & \mathbf{0} \end{pmatrix}$. Its permutation is $(n+1, \dots, 2n, 1, \dots, n)$: precisely the n^2 cross-block pairs are inversions. Hence $\det(\mathbf{P}) = (-1)^{n^2} = (-1)^n$. Since

$$\mathbf{K} = \begin{pmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{C} \end{pmatrix} \mathbf{P},$$

$$\det(\mathbf{K}) = (-1)^n \det(\mathbf{B}) \det(\mathbf{C}).$$



Exercise 4.4 (A bordered determinant without an inverse). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and $c \in \mathbb{R}$, prove

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{y} \\ \mathbf{x}' & c \end{pmatrix} = c \det(\mathbf{A}) - \mathbf{x}' \operatorname{adj}(\mathbf{A}) \mathbf{y}.$$

No nonsingularity assumption is permitted.

Origin: H Ex. 13.12, Eq. 13.E.1

Solution. Let $\mathbf{A}_{-i,-j}$ delete row i and column j of $\mathbf{A}$. Expansion along the last row and then along the last column of each resulting minor gives

$$\begin{aligned} \det \begin{pmatrix} \mathbf{A} & \mathbf{y} \\ \mathbf{x}' & c \end{pmatrix} &= c \det(\mathbf{A}) + \sum_{j=1}^n (-1)^{n+1+j} x_j \sum_{i=1}^n (-1)^{i+n} y_i \det(\mathbf{A}_{-i,-j}) \\ &= c \det(\mathbf{A}) - \sum_{j=1}^n \sum_{i=1}^n x_j (-1)^{i+j} \det(\mathbf{A}_{-i,-j}) y_i \\ &= c \det(\mathbf{A}) - \mathbf{x}' \operatorname{adj}(\mathbf{A}) \mathbf{y}. \end{aligned}$$



Exercise 4.5 (Schur complements under block recoding). Let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}, \quad \mathbf{A} \in \mathbb{R}^{m \times m}, \quad \mathbf{D} \in \mathbb{R}^{n \times n},$$

where $\mathbf{A}$ is nonsingular. Let $\mathbf{P}, \mathbf{R} \in \mathbb{R}^{m \times m}$ and $\mathbf{Q}, \mathbf{T} \in \mathbb{R}^{n \times n}$ be nonsingular. Define

$$\tilde{\mathbf{M}} = \begin{pmatrix} \mathbf{P} & \mathbf{O} \\ \mathbf{O} & \mathbf{Q} \end{pmatrix} \mathbf{M} \begin{pmatrix} \mathbf{R} & \mathbf{O} \\ \mathbf{O} & \mathbf{T} \end{pmatrix}.$$

Prove that the Schur complement of $\mathbf{PAR}$ in $\tilde{\mathbf{M}}$ is

$$\mathbf{Q}(\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B})\mathbf{T}.$$

Deduce that its rank equals $\text{rank}(\mathbf{S}_A)$.

Origin: Course formulation from H Thm. 8.5.10; A&M Ex. 5.10

Solution. The four blocks of $\tilde{\mathbf{M}}$ are

$$\mathbf{PAR}, \quad \mathbf{PBT}, \quad \mathbf{QCR}, \quad \mathbf{QDT}.$$

Since $(\mathbf{PAR})^{-1} = \mathbf{R}^{-1}\mathbf{A}^{-1}\mathbf{P}^{-1}$, its complement is

$$\begin{aligned} \mathbf{QDT} - \mathbf{QCR}(\mathbf{PAR})^{-1}\mathbf{PBT} \\ = \mathbf{QDT} - \mathbf{QCA}^{-1}\mathbf{BT} = \mathbf{QS}_A\mathbf{T}. \end{aligned}$$

Rank is invariant under nonsingular left and right factors. ■

Exercise 4.6 (Kernel parametrisation after block cancellation). Let $\mathbf{A} \in \mathbb{R}^{m \times m}$ be nonsingular and let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$$

have q columns in its second column block. Put $\mathbf{S} = \mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}$. Prove

$$\text{null}(\mathbf{M}) = \left\{ \begin{pmatrix} -\mathbf{A}^{-1}\mathbf{B}\mathbf{y} \\ \mathbf{y} \end{pmatrix} : \mathbf{y} \in \text{null}(\mathbf{S}) \right\}.$$

Deduce $\dim \text{null}(\mathbf{M}) = \dim \text{null}(\mathbf{S})$.

Origin: H Thm. 8.5.10, Eq. 8.5.15; A&M Ex. 5.48

Solution. For $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{y} \in \mathbb{R}^q$,

$$\begin{aligned} \mathbf{M} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \mathbf{0} &\iff \mathbf{Ax} + \mathbf{By} = \mathbf{0}, \quad \mathbf{Cx} + \mathbf{Dy} = \mathbf{0} \\ &\iff \mathbf{x} = -\mathbf{A}^{-1}\mathbf{By}, \quad (\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B})\mathbf{y} = \mathbf{0}. \end{aligned}$$

The map $\mathbf{y} \mapsto (-\mathbf{A}^{-1}\mathbf{By}, \mathbf{y})'$ is linear and one-to-one, so it is an isomorphism from $\text{null}(\mathbf{S})$ onto $\text{null}(\mathbf{M})$. 

Exercise 4.7 (Rectangular rank twins). For $\mathbf{B} \in \mathbb{R}^{m \times n}$ and $\mathbf{C} \in \mathbb{R}^{n \times m}$, prove

$$\text{rank}(\mathbf{I}_m - \mathbf{BC}) = \text{rank}(\mathbf{I}_n - \mathbf{CB}) + m - n.$$

Origin: A&M Ex. 5.50; H Thm. 8.5.10

Solution. Apply Theorem 4.6 to

$$\mathbf{K} = \begin{pmatrix} \mathbf{I}_m & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_n \end{pmatrix}$$

once with each identity block as the nonsingular diagonal block:

$$\text{rank}(\mathbf{K}) = m + \text{rank}(\mathbf{I}_n - \mathbf{CB}) = n + \text{rank}(\mathbf{I}_m - \mathbf{BC}).$$

Rearrangement gives the identity. 

Exercise 4.8 (A determinant identity with a scalar parameter). For $\mathbf{B} \in \mathbb{R}^{m \times n}$, $\mathbf{C} \in \mathbb{R}^{n \times m}$, and $t \neq 0$, prove

$$\det(t\mathbf{I}_m - \mathbf{BC}) = t^{m-n} \det(t\mathbf{I}_n - \mathbf{CB}).$$

Origin: H Cor. 18.1.2; A&M Ex. 5.37(b)

Solution. Corollary 4.10, with $\mathbf{U} = -t^{-1}\mathbf{B}$ and $\mathbf{V} = \mathbf{C}$, gives

$$\det(\mathbf{I}_m - t^{-1}\mathbf{BC}) = \det(\mathbf{I}_n - t^{-1}\mathbf{CB}).$$

Since

$$\det(t\mathbf{I}_m - \mathbf{BC}) = t^m \det(\mathbf{I}_m - t^{-1}\mathbf{BC}),$$

and the corresponding factor on the right is t^n , the stated identity follows. 

Exercise 4.9 (A block sum-and-difference identity). For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$, with no nonsingularity or commutativity assumptions, prove

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} = \det(\mathbf{A} + \mathbf{B}) \det(\mathbf{A} - \mathbf{B}).$$

Origin: A&M Ex. 5.38

Solution. Block multiplication gives

$$\begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{O} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} \\ = \begin{pmatrix} \mathbf{A} + \mathbf{B} & \mathbf{O} \\ \mathbf{O} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{O} \\ \mathbf{B} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{O} & \mathbf{A} - \mathbf{B} \end{pmatrix}.$$

The two unit block factors have determinant 1. The block-triangular determinant formula applied to the remaining factors gives the identity. 

Exercise 4.10 (A three-block arrow matrix). Let

$$\mathbf{Z} = \begin{pmatrix} \mathbf{A} & \mathbf{B} & \mathbf{C} \\ \mathbf{B}' & \mathbf{D} & \mathbf{O} \\ \mathbf{C}' & \mathbf{O} & \mathbf{E} \end{pmatrix},$$

where $\mathbf{D}$ and $\mathbf{E}$ are nonsingular square matrices. Prove

$$\begin{aligned} \det(\mathbf{Z}) &= \det(\mathbf{D}) \det(\mathbf{E}) \det(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}' - \mathbf{C}\mathbf{E}^{-1}\mathbf{C}'), \\ \text{rank}(\mathbf{Z}) &= \text{rank}(\mathbf{D}) + \text{rank}(\mathbf{E}) + \text{rank}(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}' - \mathbf{C}\mathbf{E}^{-1}\mathbf{C}'). \end{aligned}$$

Origin: A&M Ex. 5.39 and 5.52

Solution. Regroup $\mathbf{Z}$ into a 2 by 2 partition:

$$\tilde{\mathbf{B}} = (\mathbf{B} \ \mathbf{C}), \quad \tilde{\mathbf{D}} = \begin{pmatrix} \mathbf{D} & \mathbf{O} \\ \mathbf{O} & \mathbf{E} \end{pmatrix}, \quad \mathbf{Z} = \begin{pmatrix} \mathbf{A} & \tilde{\mathbf{B}} \\ \tilde{\mathbf{B}}' & \tilde{\mathbf{D}} \end{pmatrix}.$$

Then

$$\tilde{\mathbf{D}}^{-1} = \begin{pmatrix} \mathbf{D}^{-1} & \mathbf{O} \\ \mathbf{O} & \mathbf{E}^{-1} \end{pmatrix}, \quad \det(\tilde{\mathbf{D}}) = \det(\mathbf{D}) \det(\mathbf{E}), \quad \text{rank}(\tilde{\mathbf{D}}) = \text{rank}(\mathbf{D}) + \text{rank}(\mathbf{E}),$$

and

$$\mathbf{A} - \tilde{\mathbf{B}}\tilde{\mathbf{D}}^{-1}\tilde{\mathbf{B}}' = \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}' - \mathbf{C}\mathbf{E}^{-1}\mathbf{C}'.$$

Apply Theorems 4.9 and 4.6 with the $\tilde{\mathbf{D}}$ as the nonsingular diagonal block. ■

Exercise 4.11 (An updated block column). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be nonsingular, $\mathbf{U} \in \mathbb{R}^{n \times k}$, $\mathbf{C} \in \mathbb{R}^{k \times k}$, and $\mathbf{V} \in \mathbb{R}^{k \times n}$. Suppose $\mathbf{I}_k + \mathbf{CVA}^{-1}\mathbf{U}$ is nonsingular. Prove

$$(\mathbf{A} + \mathbf{UCV})^{-1}\mathbf{U} = \mathbf{A}^{-1}\mathbf{U}(\mathbf{I}_k + \mathbf{CVA}^{-1}\mathbf{U})^{-1}.$$

Origin: Course formulation from H Cor. 18.2.9 and Thm. 18.2.8

Solution. Set $\mathbf{K} = \mathbf{I}_k + \mathbf{CVA}^{-1}\mathbf{U}$. Then

$$(\mathbf{A} + \mathbf{UCV})\mathbf{A}^{-1}\mathbf{U} = \mathbf{U} + \mathbf{UCVA}^{-1}\mathbf{U} = \mathbf{UK}.$$

Multiplicativity and Corollary 4.10 give

$$\det(\mathbf{A} + \mathbf{UCV}) = \det(\mathbf{A}) \det(\mathbf{I}_k + \mathbf{CVA}^{-1}\mathbf{U}) \neq 0,$$

without requiring $\mathbf{C}$ to be nonsingular. Premultiplication by $(\mathbf{A} + \mathbf{UCV})^{-1}$ and postmultiplication by $\mathbf{K}^{-1}$ yield the formula. ■

Exercise 4.12 (Equivalent determinant updates). Let $\mathbf{R} \in \mathbb{R}^{n \times n}$ be nonsingular, $\mathbf{S} \in \mathbb{R}^{n \times m}$, $\mathbf{T} \in \mathbb{R}^{m \times m}$, and $\mathbf{U} \in \mathbb{R}^{m \times n}$. Prove, without assuming $\mathbf{T}$ nonsingular, that

$$\det(\mathbf{R} + \mathbf{STU}) = \det(\mathbf{R}) \det(\mathbf{I}_m + \mathbf{UR}^{-1}\mathbf{ST}) = \det(\mathbf{R}) \det(\mathbf{I}_m + \mathbf{TUR}^{-1}\mathbf{S}).$$

Origin: H Ex. 18.2

Solution. Multiplicativity and Corollary 4.10 give

$$\begin{aligned} \det(\mathbf{R} + \mathbf{STU}) &= \det(\mathbf{R}) \det(\mathbf{I}_n + \mathbf{R}^{-1}\mathbf{STU}) \\ &= \det(\mathbf{R}) \det(\mathbf{I}_m + \mathbf{UR}^{-1}\mathbf{ST}). \end{aligned}$$

A second application of that corollary, now to $\mathbf{UR}^{-1}\mathbf{S}$ and $\mathbf{T}$, gives

$$\det(\mathbf{I}_m + \mathbf{UR}^{-1}\mathbf{ST}) = \det(\mathbf{I}_m + \mathbf{TUR}^{-1}\mathbf{S}).$$



Exercise 4.13 (The null direction of a rank-one update). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be nonsingular and let nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Prove that

$$\mathbf{A} + \mathbf{uv}' \text{ is singular} \iff 1 + \mathbf{v}'\mathbf{A}^{-1}\mathbf{u} = 0.$$

In the singular case, prove

$$(\mathbf{A} + \mathbf{uv}')\mathbf{A}^{-1}\mathbf{u} = \mathbf{0}, \quad \text{rank}(\mathbf{A} + \mathbf{uv}') = n - 1.$$

Origin: H Cor. 18.2.10; S Prop. 3.21

Solution. The equivalence is Corollary 4.14. If its scalar condition vanishes, then

$$(\mathbf{A} + \mathbf{uv}')\mathbf{A}^{-1}\mathbf{u} = \mathbf{u} + \mathbf{uv}'\mathbf{A}^{-1}\mathbf{u} = \mathbf{u}(1 + \mathbf{v}'\mathbf{A}^{-1}\mathbf{u}) = \mathbf{0}.$$

Since $\mathbf{A}^{-1}\mathbf{u} \neq \mathbf{0}$, the update is singular. The rank inequality for a sum gives

$$n = \text{rank}(\mathbf{A}) = \text{rank}((\mathbf{A} + \mathbf{u}\mathbf{v}') - \mathbf{u}\mathbf{v}') \leq \text{rank}(\mathbf{A} + \mathbf{u}\mathbf{v}') + \text{rank}(\mathbf{u}\mathbf{v}') = \text{rank}(\mathbf{A} + \mathbf{u}\mathbf{v}') + 1.$$

Singularity gives $\text{rank}(\mathbf{A} + \mathbf{u}\mathbf{v}') < n$; hence its rank is $n - 1$. ■

Exercise 4.14 (A rank-one update without chosen factors). Let $\mathbf{R} \in \mathbb{R}^{n \times n}$ be nonsingular and let $\mathbf{B} \in \mathbb{R}^{n \times n}$ have rank one. Prove

$$\mathbf{R} + \mathbf{B} \text{ is nonsingular} \iff 1 + \text{tr}(\mathbf{R}^{-1}\mathbf{B}) \neq 0,$$

and, in that case,

$$(\mathbf{R} + \mathbf{B})^{-1} = \mathbf{R}^{-1} - \frac{\mathbf{R}^{-1}\mathbf{B}\mathbf{R}^{-1}}{1 + \text{tr}(\mathbf{R}^{-1}\mathbf{B})}.$$

Origin: H Ex. 18.8(a)

Solution. The rank-one expansion theorem gives $\mathbf{B} = \mathbf{u}\mathbf{v}'$ for nonzero $\mathbf{u}, \mathbf{v}$. Componentwise,

$$\operatorname{tr}(\mathbf{R}^{-1}\mathbf{B}) = \operatorname{tr}((\mathbf{R}^{-1}\mathbf{u})\mathbf{v}') = \sum_{i=1}^n (\mathbf{R}^{-1}\mathbf{u})_i v_i = \mathbf{v}'\mathbf{R}^{-1}\mathbf{u}.$$

Substitute $\mathbf{B} = \mathbf{u}\mathbf{v}'$ into Corollary 4.14. 